

Feb 2025

Roll No. 22ENAC5028

Total Page No. : 3

11N502

11N502

**B.TECH. I SEMESTER (NEW SCHEME)
MAIN/BACK EXAMINATION 2024-25**

ALL BRANCH

1FY1-02 - Engineering Physics

Time : 3 Hours]

[Max. Marks : 70

[Min. Passing Marks :

Instructions to Candidates :

Part-A : Short Answer Type Questions (up to 25 words) $10 \times 2 = 20$ marks. All 10 questions are compulsory.

Part-B : Analytical/Problem Solving questions $5 \times 4 = 20$ marks. Candidates have to answer 5 questions out of 7.

Part-C : Descriptive/Analytical/Problem Solving questions 3×10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of the following supporting materials is permitted during examination.
(Mentioned in form no. 205).

1 _____

2 _____

Part-A

Q. 1. Why is the centre dark in Newton's ring experiment.

[2]

Q. 2. Draw neat and well labelled diagram of grating spectrum.

[2]

B-325

(1)

P.T.O.

- Q. 3. Write any *two* importance of Quantum Mechanics. [2]
- Q. 4. What is a wave-function ? Give its physical significance. [2]
- Q. 5. Differentiate between spontaneous and stimulated emissions with suitable diagrams. [2]
- Q. 6. Define Numerical Aperture of an optical fibre. Also, write its formula. [2]
- Q. 7. Mention any *two* differences between type-I and type-II super conductors. [2]
- Q. 8. Write any *two* applications of Nanotechnology. [2]
- Q. 9. Define curl of a vector quantity. Also give its physical interpretation. [2]
- Q. 10. What do you mean by displacement current ? Give its formula. [2]

Part-B

- Q. 1. Derive an expression for wavelength determination by Michelson Interferrometer with the help of a neat and well-labelled diagram. [4]
- Q. 2. How many lines per cm are there in a grating which gives an angle of diffraction of 30° in first order spectrum of light of wavelength 6×10^{-5} cm. [4]
- Q. 3. Derive Schrödinger time-independent wave equation in one-dimension. [4]
- Q. 4. For an optical fibre with core glass of refractive index 1.5 and fractional index difference of 0.0005. Determine : [4]
- (i) The cladding index
 - (ii) Critical internal reflection angle Q_c
 - (iii) Numerical Aperture
- Q. 5. Explain top-down and bottom-up approach for synthesis of nanomaterials with suitable diagram. [4]
- Q. 6. State and derive Bragg's law for X-ray diffraction. [4]
- Q. 7. State and prove poynting theorem. [4]

Part-C

- Q. 1. What will be the effect on Newton's rings if :
- A plane mirror is used instead of the glass plate below the plano convex lens.
 - A plano convex lens of small radius of curvature is used.
 - If the plano convex lens is raised by height Δh from the surface of plane glass plate. [10]
- Q. 2. Obtain an expression for energy and wave-function of a particle confined in a 1D-Box of size 'a'. Show that the energy of this particle is discrete. [10]
- Q. 3. With the help of a well-labelled and neat diagram explain the construction and working of He : Ne LASER. [10]
- Q. 4. Distinguish between different type of semiconductors. Derive an expression for the electrical conductivity of semiconductors. [10]
- Q. 5. Derive Maxwell's fourth equation on electromagnetism. State the necessary modification made by Maxwell. [10]

Q4

Ans

$$n_{\text{core}} = 1.5 \quad \Delta = 0.0005$$

$$N \cdot A = n_{\text{core}} \sqrt{2\Delta}$$

$$\textcircled{1} \quad N \cdot A = 1.5 \sqrt{2 \times 0.0005} \quad \text{Ans.}$$

$$\textcircled{11} \quad \sin \theta_c = \frac{n_{\text{cladd.}}}{n_{\text{core.}}}$$

$$\textcircled{11} \quad D = \frac{n_{\text{core}} - n_{\text{cladd.}}}{n_{\text{core.}}}$$

B-325

(3)

$$0.0005 = \frac{1.5 - n_{\text{cladd.}}}{1.5}$$

$$n_{\text{cladd.}} = 1.49 \quad \text{Ans}$$

$$\Delta = \frac{n_{\text{core}} - n_{\text{cladd.}}}{n_{\text{core.}}}$$

Ex.12 How many lines per cm are there in a grating which gives an angle of diffraction of 30° in first order spectrum of light of wavelength 6×10^{-5} cm?

[RTU Jan., 2003]

Sol. The grating equation is

$$(e + b) \sin \theta = n\lambda$$

$$\Rightarrow (e + b) = \frac{n\lambda}{\sin \theta} = \frac{1 \times 6 \times 10^{-5}}{\sin 30^\circ} = 12 \times 10^{-5} \text{ cm}$$

$$\text{No. of lines per cm} = \frac{1}{(e + b)} = \frac{10^5}{12} = 8333.3/\text{cm}$$

Ans.

A-054

Roll No. 23FLDME008

Total No. of Pages : 03

21N502 /

B.Tech. II Sem Main/Back (New Scheme) Acad. Session 2023-24

All Branch

(2FY1-02) – Engineering Physics

21N502/

Time : 3 Hours

Maximum Marks: 70

Min. Passing Marks:

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Part – A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All 10 questions are compulsory.

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Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

P.T.O.

PART-A

- Q.1. What is thin-film Interference. [2]
- Q.2. In fraunhofer diffraction, what kind of source of light is used and where it is situated? [2]
- Q.3. During Population inversion, which process is dominant? [2]
- Q.4. What is the NA of the coaxial cable with its core and cladding indexes as 2.44 & 1.2? [2]
- Q.5. What is quantum mechanical tunneling? [2]
- Q.6. What is Wave function? [2]
- Q.7. Define energy bands in Solids. [2]
- Q.8. Write two applications of nano materials. $\frac{1}{2} \epsilon_0 E^2$ [2]
- Q.9. In a wave if $E_0 = 100$ V/m. Then find the magnitude of poynting vector. [2]
- Q.10. What is displacement current & its formula? $\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ $= \frac{I}{A}$

PART-B

- Q.1. In a Newton's ring experiment, the diameter of the 20th dark ring was found to be 5.82 mm and the 10th ring is 3.36 mm. If the radius of the plano convex lens is 1m, calculate the Wavelength of light used. [4]
- Q.2. What are Einstein coefficients? Derive the expression for the relation between Einstein coefficients. [1+3=4]
- Q.3. For $n^2 x + n^2 y + n_z^2 = 26$ calculate the degeneracy, write the wave functions for the degenerated states and energy levels of degenerated states. [4]
- Q.4. A silver ribbon whose cross section is 1.0 cm by 0.2 cm. The ribbon carries a current of 100 A from left to right, and it lies in a uniform magnetic field of magnitude 1.5T. Using a density value $n = 5.9 \times 10^{28}$ electrons per cubic meter of silver, find the hall potential between the edges of the ribbon. [4]

$$V_H = \frac{IB}{nqt} = \frac{100 \text{ A} \times 1.5 \text{ T}}{5.9 \times 10^{28} \times 1.6 \times 10^{-19} \times \frac{1.0}{0.2}}$$

Ans

$$\phi_B = B \cdot A \cdot \cos \theta$$

magnetic flux

- Q.5. A circular antenna of area 3m^2 is installed at a place in Madurai. The plane of the area of antenna is inclined at 47° with the direction of Earth's magnetic field. If the magnitude of Earth's field at the place is $40773.9 \times 10^{-9}\text{T}$.

Find the magnetic flux linked with the antenna.

[4]

- Q.6. Derive the expression for time dependent Schrodinger wave equation. [4]

- Q.7. Draw the line diagram of Michelson's Interferometer and explain its construction.

[4]

PART- C

- Q.1. Derive the expression of the intensity of light due to diffraction by a single slit.

[10]

- Q.2. Describe the structure, types and features of optical fiber.

[4+4+2]

- Q.3. Derive the expression of wave function and energy for the particle in a one dimensional potential box.

[10]

- Q.4. Describe the properties of nano materials. Explain the synthesis of nanomaterials using top-down and bottom-up approach.

[2+8]

- Q.5. Derive the maxwell equations in their integral form?

[10]

$$N \cdot A = \frac{\sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2}}{h_0} = \frac{\sqrt{(2.44)^2 - (1.2)^2}}{1}$$

$$N \cdot A = \sqrt{5.9536 - 1.44} = \sqrt{4.5136} \text{ Ans}$$

Q9

Ans. Power $\rightarrow P = \frac{1}{2} \epsilon_0 |E|^2$

given electric field Amplitude $E_0 = 100 \text{ V/m}$.

$$\epsilon_0 = 8.85 \times 10^{-12}$$

$$P = \frac{1}{2} 8.85 \times 10^{-12} \times (100)^2 \quad \underline{\underline{\text{Ans.}}}$$

Q3

Ans. $n_x^2 + n_y^2 + n_z^2 = 26$

$$4^2 + 3^2 + 1^2 = 26$$

$$16 + 9 + 1 = 26$$

Degeneracy is $(4,3,1) (4,1,3) (3,4,1) (3,1,4)$
 $(1,3,4) (1,4,3)$

Wavefunction $\psi_n = \sqrt{\frac{2}{a}} \sin \frac{4\pi x}{a} \sin \frac{3\pi y}{b} \sin \frac{\pi z}{c}$

Energy $E_n = \frac{n_x^2 + n_y^2 + n_z^2 \pi^2 \hbar^2}{2ma^2}$

$$E_n = \frac{26 \pi^2 \hbar^2}{2ma^2} \quad \underline{\underline{\text{Ans.}}}$$

Q8]

$$\lambda = \frac{D_{20}^2 - D_{10}^2}{4PR}$$

$$P = 10$$

$$R = 1 \text{ m}$$

Roll No. 21FL0CS037 **21N502 /**

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B.Tech. II Sem (New Scheme) Main Exam Acad Session 2022-23
All Branch

(2FY1-02) - Engineering Physics

21N502 /

Time : 3 Hours

Maximum Marks: 70
Min. Passing Marks:

Instructions to Candidates:

Part – A: Short answer questions (up to 25 words) $10 * 2$ marks = 20 marks. All 10 questions are compulsory.

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Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

PART-A

- Q.1. Write the ratio of the diameter of dark Newton rings. [2]
Q.2. In a single slit experiment write the formula for the intensity of secondary maxima. [2]
Q.3. Write the Plank radiation formula for energy density. [2]
Q.4. Draw the Basic structure of optical fibre. [2]
Q.5. In compton effect write the formula for shift in wavelength. [2]
Q.6. Explain normalization of wave function. [2]
Q.7. Explain, what is fermi level? [2]
Q.8. Explain, what is Meissner E effect? [2]

P.T.O.

Q.9. Explain the physical interpretation of curl. [2]

Q.10. Explain the displacement current. [2]

PART-B

Q.1. In Newton's ring experiment the diameter of n^{th} and $(n+1)^{\text{th}}$ rings are 4.2 mm and 7.0 mm. If the radius of curvature of lens is 2m, then find the wavelength used. [4]

Q.2. Light of wavelength $6000 \text{ \AA}$ is incident normally on a slit of width $24 \times 10^{-5} \text{ cm}$. Determine the angular position of second minimum. [4]

Q.3. Calculate the refractive indices of the core and cladding materials of a fiber from the data. $\text{NA} = 0.22$ and $\Delta = 0.012$. [4]

Q.4. Write a short note on semiconductor laser. [4]

Q.5. An electron is confined to a box of length 10^{-9} m . Calculate the minimum uncertainty in its velocity. [4]

Q.6. The hall coefficient of certain silicon specimens found to be $-7.5 \times 10^{-5} \text{ m}^3/\text{C}$ at a certain temperature. If the conductivity is found to be $200 \Omega^{-1} \text{ m}^{-1}$. Calculate the density of charge carriers. [4]

Q.7. Find $\nabla \cdot V$ for the following vector fields: [4]

a. $V = (y^2 + z^2)(x + y)\hat{x} + (z^2 + x^2)(y + z)\hat{y} + (x^2 + y^2)(z + x)\hat{z}$

b. $V = f_1(y, z)\hat{x} + f_2(x, z)\hat{y} + f_3(x, y)\hat{z}$

PART-C

Q.1. Describe the construction and working of Michelson Interferometer. Drive the expression for determine the wavelength difference of two components of a line by Michelson Interferometer. [10]

Q.2. Explain the term absorption, spontaneous emission and stimulated emission and derive a relation between Einstein's coefficients. [10]

- Q.3. Write down schrodinger's equation for a particle confined in a 1-D box. Obtain the wave function for a particle confined in this box. [10]
- Q.4. Describe the formation of energy band in solids and hence how it helps to classify the materials into conductors and insulators. [10]
- Q.5. Deduce the Maxwell's equations for free space and prove that electromagnetic waves are transverse. [10]

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Roll No. _____

Total No. of Pages: 3

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B. Tech. I - Sem. (New Scheme) (Main) Exam., May - 2023

All Branch

1FY1 - 02 Engineering Physics

Common to all Branches

Time: 3 Hours

Maximum Marks: 70

Instructions to Candidates:

Part - A: Short answer questions (up to 25 words) 10×2 marks = 20 marks.
All ten questions are compulsory.

Part - B: Analytical/Problem solving questions 5×4 marks = 20 marks.
Candidates have to answer **five** questions out of **seven**.

Part - C: Descriptive/Analytical/Problem Solving/Design questions 3×10 marks = 30 marks. Candidates have to answer **three** questions out of **five**.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination.
(Mentioned in form No. 205)

1. NIL

2. NIL

PART - A

- Q.1 State the essential difference between interference and diffraction of light. [2]
- Q.2 What is physical significance of wave function? [2]
- Q.3 The uncertainty principle imposes no restriction on the measurements related to macroscopic objects. Comment. [2]
- Q.4 Explain how X-ray diffraction can be used to characterize nano particles? [2]

- Q.5 / Sketch a neat diagram of Fraunhofer diffraction at a single slit. [2]
- Q.6 / State the Maxwell's equations in their differential form. [2]
- Q.7 Explain the terms - [2]
- (i) Basis
 - (ii) Space lattice
 - (iii) Unit cell
 - (iv) Space group
- Q.8 Write a short note on Davisson Germer's experiment. [2]
- Q.9 / Write two examples of type-I and type-II superconductors. [2]
- Q.10 / What is meant by numerical aperture for an optical fibre? [2]

PART - B

- Q.1 Explain how you can distinguish between linearly, circularly and elliptically polarized light. [4]
- Q.2 / Explain how Newton's rings experiment can be used to find the wavelength of the light source used. [4]
- Q.3 What is de Broglie's hypothesis? Calculate the de-Broglie wavelength associated with electrons, which are accelerated by a voltage of 50 kV. [4]
- Q.4 Explain top down and bottom up approach for the synthesis of nano materials. [4]
- Q.5 Calculate the Zero-point energy for a particle in an infinite potential well for an electron confined to a 1 nm atom. [4]
- Q.6 Explain the principle and working of a He-Ne laser by clearly drawing the energy level diagrams. [4]
- Q.7 / What is Hall effect and its importance in semiconductor industry? [4]

PART - C

- Q.1 What are orthogonal wave functions? Write the Schrodinger equation for particle in a box and solve it to obtain energy Eigen values and Eigen functions. [10]
- Q.2/ (a) Explain what is population inversion and pumping in lasers? [4]
(b) With neat diagrams, describe the principle, construction and working of semiconductor laser. Also, discuss the application of lasers. [6]
- Q.3 (a) What is diffraction grating? Describe how the wavelength of the monochromatic light is determined using it. [5]
(b) Find the number of orders visible if the wavelength of the incident radiation is $5000 \text{ \AA}$ and number of lines on the grating are 1000 in one centimeter. [5]
- Q.4 (a) How would you produce polarized light by reflection? What is Brewster's Law? Calculate the angular position of the sun above the horizon so that light reflected from a calm lake is completely polarized. The refractive index of water is 1.33. [5]
(b) How would you ascertain the correct state of polarization (i.e. circularly, elliptically, plane, unpolarized, partial polarized etc.) of an unknown incoming beam of light with the help of a rotating Nicol prism and a quarter wave plate? [5]
- Q.5 (a) Why semiconductors are doped? Giving suitable energy level diagram explain how doping by donors improve conductivity of a semiconductor. [7]
(b) Explain Meissner's effect. [3]
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11N502

11N502

B.TECH. I SEM (NEW SCHEME) MAIN/BACK

2023-24

ENGINEERING PHYSICS

1FY1-02 - Common to All

Time : 3 Hours]

[Max. Marks : 70

Instructions to Candidates:

Part-A : Short Answer Type Questions (up to 25 words) 10×2 marks = 20 marks. All 10 questions are compulsory.

Part-B : Analytical/Problem Solving questions 5×4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

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Use of the following supporting materials is permitted during examination. (Mentioned in form no. 205).

1 _____

2 _____

Part-A $2 \times 10 = 20$ **(Short Answer Type Questions)**

1. Draw a well-labelled diagram of Newton's ring experiment.
2. How does a grating resolve spectral lines ? Explain briefly.
3. What is Heisenberg Uncertainty principle ?
4. What do you mean by Tunnel Effect ? Write two of its applications.

5. What is the full form of LASER ? Write any two of its properties.
6. Write any *two* applications of an optical fibre.
7. What do you mean by a 'Band' ? Write its type with proper distinguishment.
8. Distinguish between bottom-up and top-down approach for fabrication of nanomaterials with one example of each.
9. What do you mean by divergence of a vector quantity ? Give its physical significance.
10. Derive Maxwell's I equation of electro magnetism.

Part-B

4×5=20

1. Explain interference in thin films and also derive an expression for path difference between reflected rays.
2. Explain formation of grating spectrum with the help of a well-labelled diagram.
3. Find the probability that a particle in a box L wide can be found between $x = 0$ and $x = L/n$ when it is in the n th state.
4. Derive Einstein relations between the coefficients of absorption, spontaneous and stimulated emission of radiation.
5. Distinguish between conductors, semiconductors and insulators on the basis of band theory.
6. What is superconductivity ? Explain Meissner effect in super conductors. Also, write two of its applications.
7. Verify that vector $\hat{A} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ is both solenoidal and rotational.

Part-C

3×10=30

1. Show that the relative intensities of successive maxima of Fraunhofer diffraction at single slit are nearly :

$$I_0 : \frac{4I_0}{9\pi^2} : \frac{4I_0}{25\pi^2} : \dots\dots\dots$$

2. Solve time independent Schrödinger equation for a particle confined in a 3D cubical box of sides 'a'. Also, explain degeneracy of energy levels.
3. Explain the construction and working of semiconductor LASER. How population inversion is achieved in this laser ?
4. What is Hall effect ? Derive an expression for Hall coefficient. Write any *two* uses of this experiment.
5. Define Poynting vector. Also state and prove Poynting theorem.

A-054

Roll No. 21ELDCE010

Total No. of Pages.: 03

21N502 /

B.Tech. II Sem Main/Back (New Scheme) Acad. Session 2023-24

All Branch

(2FY1-02) – Engineering Physics

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1 _____

2 _____

A-054

PART-A

- Q.1. What is thin-film Interference. [2]
- Q.2. In fraunhofer diffraction, what kind of source of light is used and where it is situated? [2]
- Q.3. During Population inversion, which process is dominant? [2]
- Q.4. What is the NA of the coaxial cable with its core and cladding indexes as 2.44 & 1.2? [2]
- Q.5. What is quantum mechanical tunneling? [2]
- Q.6. What is Wave function? [2]
- Q.7. Define energy bands in Solids. [2]
- Q.8. Write two applications of nano materials. [2]
- Q.9. In a wave if $E_0 = 100$ V/m. Then find the magnitude of poynting vector. [2]
- Q.10. What is displacement current & its formula? [2]

PART-B

- Q.1. In a Newton's ring experiment, the diameter of the 20th dark ring was found to be 5.82 mm and the 10th ring is 3.36 mm. If the radius of the plano convex lens is 1m, calculate the Wavelength of light used. [4]
- Q.2. What are Einstein coefficients? Derive the expression for the relation between Einstein coefficients. [1+3=4]
- Q.3. For $n^2x + n^2y + n_z^2 = 26$ calculate the degeneracy, write the wave functions for the degenerated states and energy levels of degenerated states. [4]
- Q.4. A silver ribbon whose cross section is 1.0 cm by 0.2 cm. The ribbon carries a current of 100 A from left to right, and it lies in a uniform magnetic field of magnitude 1.5T. Using a density value $n = 5.9 \times 10^{28}$ electrons per cubic meter of silver, find the hall potential between the edges of the ribbon. [4]

Q.5. A circular antenna of area 3m^2 is installed at a place in Madurai. The plane of the area of antenna is inclined at 47° with the direction of Earth's magnetic field. If the magnitude of Earth's field at the place is $40773.9 \times 10^{-9}\text{T}$.

Find the magnetic flux linked with the antenna. [4]

Q.6. Derive the expression for time dependent Schrodinger wave equation. [4]

Q.7. Draw the line diagram of Michelson's Interferometer and explain its construction.

[4]

PART- C

Q.1. Derive the expression of the intensity of light due to diffraction by a single slit.

[10]

Q.2. Describe the structure, types and features of optical fiber.

[4+4+2]

Q.3. Derive the expression of wave function and energy for the particle in a one dimensional potential box.

[10]

Q.4. Describe the properties of nano materials. Explain the synthesis of nanomaterials using top-down and bottom-up approach.

[2+8]

Q.5. Derive the maxwell equations in their integral form?

[10]

June 2025

R-117

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All Branch

(2FY1-02) – Engineering Physics

21N502

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P.T.O.

R-117

Part-A

- Q.1. Explain thin film interference. [2]
- Q.2. What is diffraction of light? Explain its two types. [2]
- Q.3. Why population inversion cannot be achieved in two energy level system? [2]
- Q.4. A fiber made of silicon within a diameter of the core is large such that it consists of core and claddings refractive indexes of 1.40 and 1.37. Find the NA of the fiber. [2]
- Q.5. Write the applications of quantum mechanical tunnelling. [2]
- Q.6. How is time-dependent schrodinger equation represented? [2]
- Q.7. How does energy bands affects the conductivity in solids? Explain. [2]
- Q.8. What are the applications of nanomaterials? [2]
- Q.9. In a EM wave if $B_0 = 100$ Tesla. Then find the magnitude of poynting vector: [2]
- Q.10. What is the difference between displacement current and actual current. [2]

PART-B

- Q.1. In Newton's ring experiment the radii of the m^{th} and $(m+4)^{\text{th}}$ dark rings are respectively $\sqrt{5}$ mm and $\sqrt{7}$ mm. What is the value of m ? [4]
- Q.2. Derive the expression for the relation between Einstein Coefficients. [4]
- Q.3. For $nx^2 + ny^2 + nz^2 = 194$ calculate the degeneracy, write the wave functions for degenerated states. [4]
- Q.4. A hall probe consists of a copper strip, $n = 8.5 \times 10^{28}$ electrons per cubic meter, which is 2.0 cm wide and 0.10 cm thick. What is the magnetic field When $I = 50\text{A}$ and the Hall potential is 4.0 MV? [4]
- Q.5. An airplane is travelling west at a speed of 1500 km/h. What is the voltage difference developed between the ends of the wings having a span of 32 m, If earth's magnetic field at the location has a magnitude of 5×10^{-4} T and the dip angle is 60° . [4]
- Q.6. Describe the gradient, divergence and curl. [4]
- Q.7. Draw the line diagram of Michel Son's inter-ferometer and explain its construction. [4]

PART- C

- Q.1. Derive the expression of the intensity of light due to diffraction grating. [10]
- Q.2. Draw the structural diagram of optical fiber, describe the types of optical fiber and what are the applications of optical fiber? [4+4+2]
- Q.3. Derive the expression of wave function and energy for the particle in a three dimensional potential box. [10]

P.T.O.

Q.4. Describe the properties of superconductors. What is meissner effect? also describe the Type-I and Type-II Super conductors. [2+4+4]

Q.5. Derive the maxwell equations in their differential form. [10]

B-002

Roll No. _____

Total No. of Pages.: 04

11N502

B.Tech. I Sem (New Scheme) Main/Back Exam Acd. Session 2025-26

Common to All Branches

(1FY1-02) – Engineering Physics

11N502

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Maximum Marks: 70

Min. Passing Marks:

Instructions to Candidates:

Part-A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All ten questions are compulsory.

Part- B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving/Design questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____ 2 _____

P.T.O.

B-002

[10x2=20]

Part-A

- Q.1. Why a broad source is needed to observe interference in a thin film? [2]
- Q.2. When the Slit width is exactly half of the grating element, which orders will be missing in grating spectra? [2]
- Q.3. Write any two properties of a LASER beam. [2]
- Q.4. What do you mean by graded index optical fiber? [2]
- Q.5. What factors affect the probability of tunneling? [2]
- Q.6. State Heisenberg's uncertainty principle. [2]
- Q.7. Mention two important properties of nano materials. [2]
- Q.8. What is Meissner effect? [2]
- Q.9. Why is the curl of an electrostatic field zero? [2]
- Q.10. Define displacement current and why Maxwell introduced it? [2]

PART-B

[5x4=20]

- Q.1. Newton's rings are observed normally in reflected light of wavelength of $5800\text{\AA}$. The diameter of the 8th dark ring is 0.40 cm. Find the radius of curvature of the plano convex lens and the thickness of the film. [4]
- Q.2. For a plane transmission grating with 5000 lines per cm, Find
- (a) The highest order of spectra which can be seen with light of wavelength $5000\text{\AA}$.
- (b) The longest wavelength of light for which spectrum can be obtained. [4]

Q.3. A active medium for LASER has

$$A_{ji} = 3 \times 10^7 \text{ s}^{-1} \text{ and } B_{ji} = 1.5 \times 10^{-12} \text{ m}^3 \text{J}^{-1} \text{ s}^{-2}$$

Find the spectral energy density $u(\nu)$ at which the probability of stimulated emission equals to spontaneous emission. [4]

Q.4. Using Schrodinger's equation, derive the allowed energy levels for a particle in a one-dimensional infinite potential box of width a . [4]

Q.5. An optical Fiber has refractive index of core to be 1.5 and the refractive index difference of core-cladding to be 1%. Determine numerical aperture and maximum angle of acceptance. [4]

Q.6. X-ray of wavelength $1.54 \text{ \AA}$ produce first order diffraction at angle 20° . Calculate the interplanar spacing. [4]

Q.7. If $\vec{A} = x^2 \hat{i} - 2y^3 \hat{j} + xy^2 \hat{k}$, find the value of

(a) $\text{div} \vec{A}$ and

(b) $\text{curl} \vec{A}$ at the point $(2, 2, 2)$. [4]

PART- C

[3x10=30]

- Q.1. Explain the construction and working of Michelson interferometer. Discuss how it can be used for the determination of Wavelength of Light? [10]
- Q.2. With the help of a neat and well-labelled diagram, explain the construction and working of Semiconductor Laser. [10]
- Q.3. Obtain an expression for energy eigen values and eigen functions of a particle confined in a 3D- Box of size 'a'. [10]
- Q.4. Discuss the theory of Hall effect and derive the expression for Hall coefficient. [10]
- Q.5. Derive Maxwell's equations in differential form and outline their physical significance. [10]
